Supplementary Material for "A holistic assessment of the horticultural fruit and nut value chains in Afghanistan: identifying key constraints and priority interventions"

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Value chain performance across crops and value chain stages

Fruit value chain performance varied considerably at both the indicator and stage levels. Overall, median expert responses at the indicator level across crops ranged from 0 to 60%. Detailed expert estimates regarding each indicator and stage for all crops are presented in Table S1.

Table S1. Summary of median expert response data with interquartile range (Q1–Q3) across fruit crops and indicators

Value chain stages	No	Indicators	Pomegranate	Apple	Cherry	Citrus (oranges)	Grape	Peach	Plum	Apricot	Almond	Fig	Pine nut	Pistachio	Walnut	Indicator Median	Stage Median (%)	Interquartile Range (IQR)	
																		Q1	Q3
Orchard establishment	1	Intensive orchard cultivation	72	88	58	58	88	88	72	58	58	30	2	21	12	**58**	**58**	**37**	**58**
	2	Orchards with certified nurseries	35	58	21	44	21	44	30	44	30	2	2	2	8	**30**	**58**	**37**	**58**
	3	Certified nursery annual sufficiency	58	58	30	88	58	58	58	58	58	16	2	12	12	**58**	**58**	**37**	**58**
	4	Breeding program (although limited in scope and funding)	17	42	17	25	25	42	33	58	50	8	8	8	8	**25**	**58**	**37**	**58**
	5	Orchards with standard design	58	58	58	44	44	58	58	58	58	21	2	12	12	**58**	**58**	**37**	**58**
	6	Orchards with proper planting distance	44	58	58	58	58	58	58	58	58	30	2	12	12	**58**	**58**	**37**	**58**
Soil and irrigation	7	Farmers performing soil tests	2	2	2	2	2	2	2	2	2	2	2	2	2	**2**	**30**	**12**	**32**
	8	Soil fertilization/year	58	58	30	58	58	58	30	30	44	30	12	21	8	**30**	**30**	**12**	**32**
	9	Recommended fertilizer dosage	30	30	21	30	30	21	21	30	21	21	2	2	2	**21**	**30**	**12**	**32**
	10	Orchards properly leveled	30	44	30	30	21	30	30	30	44	21	2	2	12	**30**	**30**	**12**	**32**
	11	Orchards receiving sufficient irrigation	58	58	58	58	58	58	44	58	44	58	44	21	30	**58**	**30**	**12**	**32**
	12	Modern vs. conventional irrigation (flood, basin, furrow)	0	0	0	8	0	0	8	8	0	8	0	0	0	**0**	**30**	**12**	**32**
	13	Farmers follow a standard irrigation schedule	21	44	44	30	44	44	35	30	21	44	58	30	12	**35**	**30**	**12**	**32**
Orchard management practices	14	Practice standard training and pruning	30	58	30	58	58	44	30	44	30	30	2	2	8	**30**	**30**	**12**	**44**
	15	Farmers' knowledge of common pests and diseases	12	12	12	12	12	12	12	12	12	12	12	12	12	**12**	**30**	**12**	**44**
	16	Farmers follow proper maturity and harvest timing	58	58	58	58	58	58	58	58	58	58	58	58	58	**58**	**30**	**12**	**44**
	17	Access to standard harvesting tools	12	12	12	12	12	12	12	12	12	12	2	2	2	**12**	**30**	**12**	**44**
	18	Uniform vs. mixed harvest due to polycultivar orchards	58	58	88	58	58	58	58	58	30	58	88	30	30	**58**	**30**	**12**	**44**

Value chain stages	No	Indicators	Pomegranate	Apple	Cherry	Citrus (oranges)	Grape	Peach	Plum	Apricot	Almond	Fig	Pine nut	Pistachio	Walnut	Indicator Median	Stage Median (%)	Interquartile Range (IQR)	
																		Q1	Q3
	19	Utilization of machinery in orchards	8	8	2	8	8	2	2	8	2	2	2	2	2	**2**	**30**	**12**	**44**
	20	Farmers' knowledge of best pre- & post-harvest practices	30	30	30	30	30	30	30	30	12	21	12	12	12	**30**	**30**	**12**	**44**
Government support	21	Available & accessible subsidy programs	17	8	8	8	17	8	8	8	8	8	8	8	8	**8**	**28**	**12**	**32**
	22	Availability of credit facilities	42	42	42	42	42	42	42	42	42	42	33	33	33	**42**	**28**	**12**	**32**
	23	Expertise of extension officers (Good vs. basic knowledge)	33	42	33	25	42	42	42	42	33	25	17	25	17	**33**	**28**	**12**	**32**
	24	Active early warning for farmers vs. non-availability	25	25	25	25	25	25	25	25	17	25	8	8	8	**25**	**28**	**12**	**32**
	25	Is there a crop-specific active growers association?	8	8	8	8	17	8	8	8	8	8	8	0	0	**8**	**28**	**12**	**32**
	26	How developed is the PRODUCTION sector?	30	30	30	44	44	30	30	30	30	12	12	12	12	**30**	**28**	**12**	**32**
On-farm handling practices	27	Follow and access farm-level sorting, grading, and packing	8	12	12	12	8	12	12	12	8	8	2	2	2	**8**	**10**	**8**	**26**
	28	Uniform vs. mixed grades due to a lack of grading tools	58	30	58	58	88	30	58	58	30	58	30	58	30	**58**	**10**	**8**	**26**
	29	Practice pre-cooling for fresh fruits	2	12	12	8	12	12	12	12	8	2	8	8	8	**8**	**10**	**8**	**26**
	30	On-farm pack house availability	12	12	12	12	12	12	12	12	2	12	8	2	2	**12**	**10**	**8**	**26**
	31	Extension services for farmers	30	30	30	30	30	12	30	30	12	12	12	12	12	**30**	**10**	**8**	**26**
	32	Standardized vs. inefficient and costly local transport system	0	8	0	17	17	0	8	8	17	8	8	8	17	**8**	**10**	**8**	**26**
Off-farm processing	33	Cold storage sufficiency	12	12	12	12	12	12	12	12	12	12	12	12	12	**12**	**12**	**12**	**23**
	34	Growers linked to local markets and processing companies	30	21	30	30	30	21	21	12	12	21	12	12	12	**21**	**12**	**12**	**23**
	35	Crop-specific and operational processing centre	12	12	12	12	12	12	12	12	8	12	12	2	2	**12**	**12**	**12**	**23**
	36	Current packaging and quality standards	2	12	12	8	12	12	8	2	8	8	8	2	8	**8**	**12**	**12**	**23**
	37	Role of SMEs vs. middlemen and exporters in processing	17	17	33	17	17	42	42	42	33	25	33	25	25	**25**	**12**	**12**	**23**
	38	Post-harvest losses	25	35	25	25	25	35	35	35	15	25	15	15	15	**25**	**12**	**12**	**23**
	39	How developed is the PROCESSING sector?	12	12	12	12	21	12	12	12	12	12	12	12	12	**12**	**12**	**12**	**23**
Transportation and market access	40	Growers' awareness of market information	12	12	12	21	12	12	12	12	12	12	21	12	12	**12**	**12**	**12**	**26**
	41	Traders' awareness of market information	58	44	30	44	58	44	58	58	58	58	58	58	44	**58**	**12**	**12**	**26**
	42	Available & affordable cold transport system	8	8	8	17	8	8	8	8	8	8	8	8	8	**8**	**12**	**12**	**26**
	43	Current exports using cold chain systems	12	12	2	2	12	8	2	12	2	2	12	2	2	**2**	**12**	**12**	**26**
	44	Exports not affected by high tariffs and border challenges	12	12	12	12	2	2	2	2	12	30	30	12	30	**12**	**12**	**12**	**26**
	45	Annual total export proportion to Pakistan and India	58	58	44	21	58	58	44	58	58	58	44	58	44	**58**	**12**	**12**	**26**
	46	Export to end-user markets	30	12	44	21	30	21	12	12	21	30	12	12	12	**21**	**12**	**12**	**26**
	47	Trade requirements met by agreements (India and Pakistan)	30	30	30	12	21	12	21	12	30	12	30	30	30	**30**	**12**	**12**	**26**
	48	Market certification compliance by the private sector	30	12	12	12	12	12	12	12	12	12	30	12	12	**12**	**12**	**12**	**26**
	49	Branding and traceability requirements met	12	12	12	12	12	12	12	12	12	12	12	12	12	**12**	**12**	**12**	**26**
	50	How developed is the MARKETING sector?	30	21	12	12	21	12	12	12	30	12	30	12	12	**12**	**12**	**12**	**26**
		Crop median	**28**	**23**	**23**	**21**	**21**	**21**	**21**	**19**	**17**	**12**	**12**	**12**	**12**				

Quality standards and certification schemes

The expert response distribution showed varied responses regarding the availability and coverage of standards and certification schemes relevant to quality assurance and export markets. Across all crops, experts' familiarity with these standards and certification schemes and their reported coverage were relatively higher for fumigation, sanitary and phytosanitary (SPS) requirements and GLOBALG.A.P. The Hazard Analysis and Critical Control Point (HACCP), ISO 22000 Food Safety Management Systems and Food Safety System Certification 22000 (FSSC 22000) were commonly reported as available but with very limited coverage, alongside substantial uncertainty. There was greater uncertainty for Fairtrade, International Featured Standards (IFS), BRCGS Global Standard Food Safety, and organic certification, as responses were mostly 'I don't know', indicating limited familiarity with these schemes. Compliance with relevant food-safety and quality standards can facilitate access to international markets and enhance export competitiveness [1]. Thus, greater attention is needed to certification awareness, compliance capacity, processing standards and product quality.

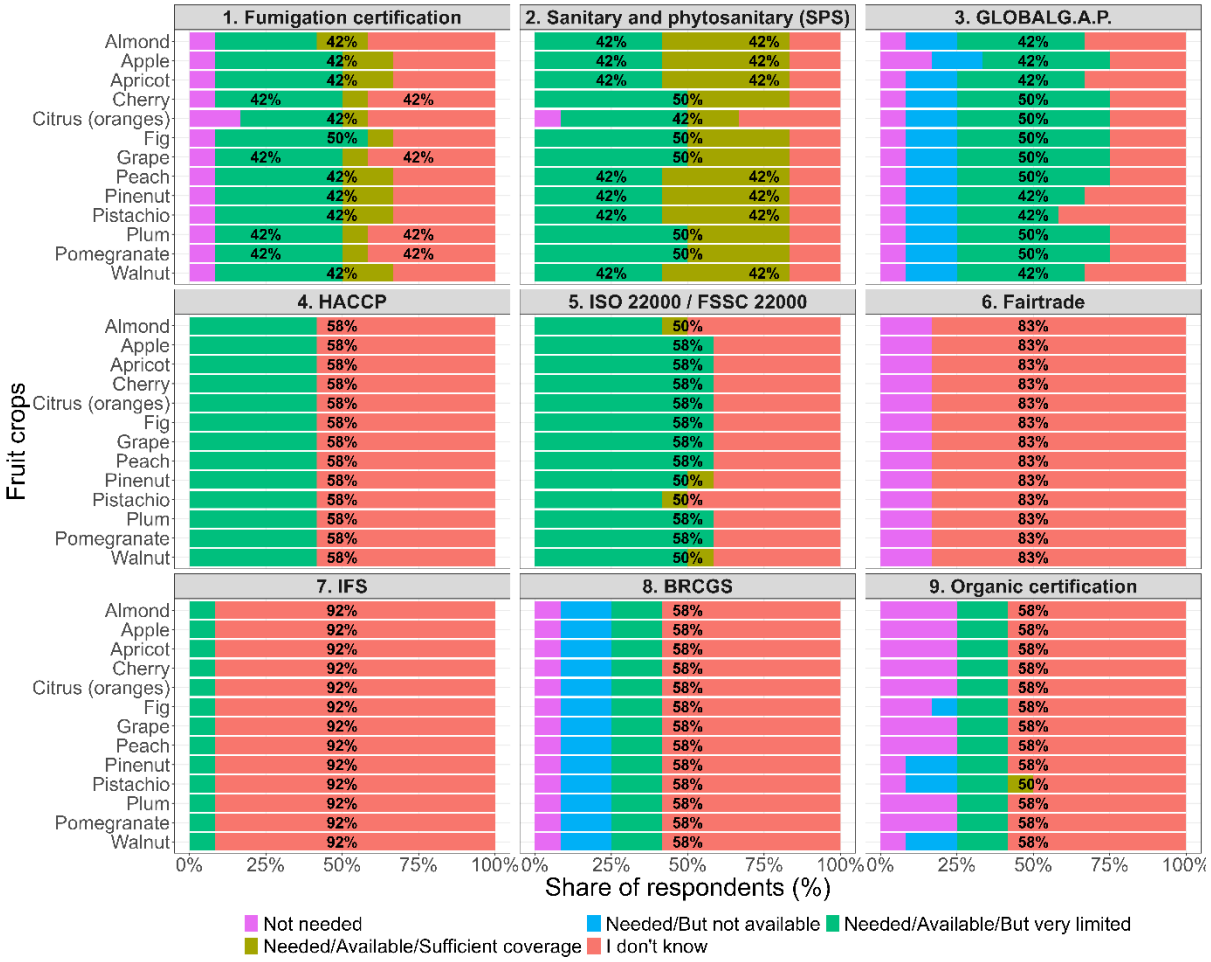


Figure S1. Expert response distributions on the need, availability and coverage of quality standards and certification schemes across fruit crops in Afghanistan.

Available literature and technical resources

Considerable work has been done to strengthen fruit value chains in recent decades. Although many challenges remain across the value chain, the existing studies provide information on best practices (Table S2). This information can help train and guide farmers in adopting these practices and also present opportunities for interventions for policymakers. In addition to full value chain studies, there are context-informed individual reports and manuals on other practices, such as establishing commercial orchards, reducing post-harvest losses, insect, pest, and disease management, harvesting and processing tools, better varieties, and export and quality standards, mostly available on the ANHDO and Roots of Peace websites. For fruits such as walnut, cherry, and mulberry, we did not identify detailed studies that could provide useful additional support.

Table S2. Fruit-specific comprehensive value chain assessment reports and guides for best practices

Crop	Document name and source	Status
Citrus	Afghanistan Citrus Value Chain Assessment, 2015	Unpublished
Apricot	Apricots in Afghanistan - A Value Chain Approach, 2008	Published, FAO
	https://www.fao.org/sustainable-food-value-chains/library/details/en/c/270959/	
Apricot	Best practices for apricot production and marketing in Afghanistan, 2016	Published, Roots of Peace
	https://roots-of-peace.squarespace.com/s/Best-Practices-for-APRICOT-Production-and-Marketing-in-Afghanistan-Roots-of-Peace-ROP.pdf	
Apple	Best practices for apple production and marketing in Afghanistan, 2016	Published, Roots of Peace
	https://roots-of-peace.squarespace.com/s/Best-Practices-for-APPLE-Production-and-Marketing-in-Afghanistan-Roots-of-Peace-ROP.pdf	
Almond	Best practices for almond production and marketing in Afghanistan, 2016	Published, Roots of Peace
	https://rootsofpeace.org/s/Best-Practices-for-ALMOND-Production-and-Marketing-in-Afghanistan.pdf	
Grape	Best practices for grape production and marketing in Afghanistan, 2016	Published, Roots of Peace
	https://rootsofpeace.org/s/Best-Practices-for-GRAPE-Production-and-Marketing-in-Afghanistan-Roots-of-Peace-ROP.pdf	
Grape	The national grape value chain, 2015-16	Unpublished, CARD-F, MAIL
Pomegranate	Best practices for pomegranate production and marketing in Afghanistan, 2016	Published, Roots of Peace
	https://rootsofpeace.org/s/Best-Practices-for-POMEGRANATE-Production-and-Marketing-in-Afghanistan-Roots-of-Peace-ROP.pdf	
Pomegranate	The national pomegranate value chain, 2017	Unpublished, MAIL

The evolving citrus value chain

Afghanistan remains a net importer of citrus, largely kinnow (mandarin), because domestic production remains low relative to demand. Estimates indicate production of approximately 14,000 tons, while

recent statistics show imports of approximately 92,000 tons [2,3]. [4] estimated the total market size at 146,000 tons, illustrating the substantial domestic demand for citrus. Thus, a significant gap remains between domestic production and total demand. Despite the domestic production deficit and reliance on imports, part of domestic citrus production is exported, particularly oranges. Afghan citrus are in high demand in other countries within the region. The value chain has been evolving, and production is increasing as new orchards are established. The country's eastern region, which exhibits a Mediterranean-type climate, is suitable for citrus production [5]. Our findings show that, despite the overall production shortfall, the value chain performs better than other fruit value chains (Figure 3 in the main manuscript) with comparatively stronger performance concentrated in the production subsector. Thus, the adoption of high-yielding varieties, together with the expansion of new orchards, may be instrumental to closing the domestic supply gap. A recent study on the suitability of different fruits using a multicriteria analysis also found that citrus, particularly sweet orange and lemon, are highly suitable for expansion due to high profit margins, market value, and local demand [6].

References

[1] Henson S, Jaffee S. Understanding Developing Country Strategic Responses to the Enhancement of Food Safety Standards. World Econ 2008; 31:548–68. https://doi.org/10.1111/j.1467-9701.2007.01034.x.

[2] FAO. Citrus fruit, total – Afghanistan 2021.

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[5] Giordani E, Cullen G, Degl'Innocenti P, Masini G. Local fruits and nuts as a tool for the development of Afghanistan. J Univ Int Dev Coop 2014:627–35.

[6] Mohammadi MH, Hejazi Z, Saeedi MA, Giordani E. Using AHP and PROMETHEE Multicriteria Decision-Making Approaches to Rank Available Fruit Crops for Orchard Expansion in Nangarhar, Afghanistan. Erwerbs-Obstbau 2023;65:1837–47. https://doi.org/10.1007/s10341-023-00902-y.